

02_Product_Requirements_Document.md

Part 3 — Non-Functional Requirements

7. Non-Functional Requirements

Purpose

Non-functional requirements define **how the system should perform**, rather than what features it provides.

These requirements are mandatory for every module of Aetheris Trade.

7.1 Performance Requirements

The platform must deliver a fast and responsive experience across all supported devices.

Page Load Time

- Landing Page < 2 seconds
 - Dashboard < 2 seconds
 - Market Page < 2.5 seconds
 - Portfolio < 2 seconds
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API Response Time

Average response:

- Under 500ms

Maximum acceptable:

- Under 1 second
-

Search Performance

Stock search should begin displaying suggestions within:

- 300 milliseconds
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Animation Performance

- 60 FPS target
 - Smooth transitions
 - No animation jank
 - GPU accelerated where possible
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Bundle Optimization

Requirements:

- Code Splitting
 - Lazy Loading
 - Route Based Loading
 - Dynamic Imports
 - Tree Shaking
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7.2 Scalability Requirements

The architecture must support future growth without major restructuring.

The system should support:

- Thousands of concurrent users
- Millions of stored trades
- Modular feature expansion
- Independent feature deployment

Architecture should remain maintainable as the product evolves.

7.3 Reliability Requirements

The application should remain stable during normal usage.

Requirements:

- Graceful error handling
 - Retry mechanisms where appropriate
 - Data consistency
 - No unexpected crashes
 - Automatic recovery from recoverable failures
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7.4 Security Requirements

Security is a core product requirement.

The platform must implement:

- Secure Authentication
- Password Hashing
- HTTPS Only
- JWT Session Security
- Input Validation
- SQL Injection Protection
- XSS Protection
- CSRF Protection
- Secure Cookies
- Rate Limiting

Sensitive information must never be exposed to the client.

7.5 Privacy Requirements

User privacy must always be respected.

Requirements:

- Collect only necessary data
 - Clear privacy policy
 - Secure storage of user information
 - No unnecessary tracking
 - User-controlled account deletion
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7.6 Accessibility Requirements

Aetheris Trade should be usable by as many people as possible.

Requirements:

- Keyboard Navigation
- Screen Reader Support
- Proper Color Contrast
- Visible Focus States
- Responsive Text Scaling
- Semantic HTML
- Accessible Forms

Target compliance:

7.7 Responsive Design Requirements

The application must work across:

- Desktop
- Laptop
- Tablet
- Mobile

Layouts should adapt without breaking functionality.

No horizontal scrolling should occur on supported devices.

7.8 Browser Support

Officially supported browsers:

- Chrome
- Edge
- Firefox
- Safari

The application should gracefully degrade on unsupported browsers.

7.9 Design Consistency

Every screen must follow the Design System.

Requirements:

- Consistent spacing
- Consistent typography
- Reusable components
- Unified color palette
- Consistent iconography
- Predictable interactions

No page should introduce a different design language.

7.10 Maintainability

The codebase should remain easy to understand and extend.

Requirements:

- Modular architecture
 - Reusable components
 - Clear folder structure
 - Type Safety
 - Documentation
 - Code comments only where necessary
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7.11 AI Requirements

Artificial Intelligence must act as an educational assistant.

The AI should:

- Explain concepts
- Analyze portfolios
- Review trades
- Teach investing
- Summarize information

The AI must not:

- Promise profits
- Guarantee returns
- Encourage reckless trading
- Present opinions as financial advice

Educational guidance must always be prioritized.

7.12 Error Handling

Every error should provide meaningful feedback.

Users should never encounter:

- Blank screens
- Unexplained failures
- Technical stack traces

Instead, provide:

- Friendly messages
 - Retry actions
 - Helpful guidance
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7.13 Logging & Monitoring

The system should log:

- Authentication events
- Trade execution
- Portfolio updates
- API failures
- AI service errors
- Unexpected exceptions

Logs should support debugging without exposing sensitive user information.

7.14 Testing Requirements

Every production release should pass:

- Unit Testing
- Integration Testing
- UI Testing
- API Testing
- Performance Testing

Critical business logic should be covered by automated tests.

7.15 Deployment Requirements

Deployment should include:

- Continuous Integration
- Continuous Deployment
- Environment Separation
- Rollback Support
- Health Checks

Production deployments should minimize downtime.

7.16 Documentation Requirements

Every module must include:

- Purpose
- Architecture
- API References
- Database Relationships
- Future Improvements

Documentation should be updated alongside implementation.

7.17 Coding Standards

Development should follow:

- Clean Architecture
- SOLID Principles
- DRY (Don't Repeat Yourself)
- KISS (Keep It Simple)
- Meaningful Naming Conventions

No unnecessary complexity should be introduced.

7.18 Product Quality Standards

Every feature must satisfy the following before release:

- Functional
 - Responsive
 - Accessible
 - Secure
 - Performant
 - Consistent with Design System
 - Reviewed
 - Tested
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Approval Criteria

This section will be considered complete when:

- All engineering quality standards are defined.
- Performance expectations are measurable.
- Security expectations are documented.

- Accessibility requirements are documented.
 - AI behavior is clearly defined.
 - Development standards are established.
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Related Documents

- 01_Project_Vision.md
 - 02_Product_Requirements_Document.md (Part 1 & Part 2)
 - 09_Design_System.md
 - 10_Development_Rules.md
 - 12_Project_Constitution.md
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Status

Version: 1.0.0

Status: Draft (Pending Review)

Next Section

Part 4 — Screen Specifications & User Flows

This section will define every screen of Aetheris Trade in detail, including its purpose, layout, components, interactions, and navigation.